

AI Assisted Coding Ass- 17.3

B.Sushanth

2403A52L11-B50

Task1:

Code and prompt:

```
week17.3 > task1.py > ...
1  """
2  Generate Python preprocessing code for a loan approval dataset using pandas and sklearn.
3  Handle missing values in loan_amount and credit_score, encode loan_status,
4  apply standard scaling, and split into train/test sets.
5  """
6  import pandas as pd
7  from sklearn.preprocessing import StandardScaler, LabelEncoder
8  from sklearn.model_selection import train_test_split
9
10 # Load dataset
11 df = pd.read_csv("loanapproval.csv")
12
13 # Handle missing values
14 df['loan_amount'].fillna(df['loan_amount'].median(), inplace=True)
15 df['credit_score'].fillna(df['credit_score'].median(), inplace=True)
16
17 # Encode target
18 le = LabelEncoder()
19 df['loan_status'] = le.fit_transform(df['loan_status']) # Approved=1, Rejected=0
20
21 # Separate features and target
22 X = df.drop('loan_status', axis=1)
23 y = df['loan_status']
24
25 # Select numeric columns for scaling
26 num_cols = X.select_dtypes(include=['int64', 'float64']).columns
27
28 scaler = StandardScaler()
29 X[num_cols] = scaler.fit_transform(X[num_cols])
30
31 # Train-test split
32 X_train, X_test, y_train, y_test = train_test_split(
33     X, y, test_size=0.2, random_state=42
34 )
35
36 print("Training shape:", X_train.shape)
```

Task2:

Code and prompt:

```

week17.3 > task2.py > ...
1  """
2  Write Python code to preprocess a social media dataset.
3  Remove duplicates, handle missing values, and encode categorical columns.
4  """
5  import pandas as pd
6
7  df = pd.read_csv("social_media.csv")
8
9  # Remove duplicates
10 df.drop_duplicates(inplace=True)
11
12 # Handle missing values
13 df.fillna(method='ffill', inplace=True)
14
15 # Encode categorical columns
16 cat_cols = df.select_dtypes(include='object').columns
17
18 for col in cat_cols:
19     df[col] = df[col].astype('category').cat.codes
20
21 print(df.head())

```

Task3:

Code and prompt:

```

week17.3 > task3.py > ...
1  """
2  Generate Python code to preprocess a weather dataset.
3  Convert date column, handle missing values,
4  create season and week_number features,
5  and normalize rainfall.
6  """
7  import pandas as pd
8  from sklearn.preprocessing import MinMaxScaler
9
10 df = pd.read_csv("weather.csv")
11
12 # Convert date
13 df['date'] = pd.to_datetime(df['date'])
14
15 # Handle missing values
16 df['temperature'].fillna(df['temperature'].mean(), inplace=True)
17 df['humidity'].fillna(df['humidity'].mean(), inplace=True)
18
19 # Feature engineering
20 df['week_number'] = df['date'].dt.isocalendar().week
21
22 def get_season(month):
23     if month in [12, 1, 2]:
24         return 'Winter'
25     elif month in [3, 4, 5]:
26         return 'Summer'
27     elif month in [6, 7, 8]:
28         return 'Monsoon'
29     else:
30         return 'Autumn'
31
32 df['season'] = df['date'].dt.month.apply(get_season)
33
34 # Normalize rainfall
35 scaler = MinMaxScaler()
36 df[['rainfall']] = scaler.fit_transform(df[['rainfall']])
37
38 print(df.head())

```

Task4:

Code and prompt:

```
week17.3 > task4.py > ...
1  """
2  Write Python preprocessing code for a movie dataset.
3  Handle missing values in rating and genre, encode categorical variables,
4  remove duplicates, and scale ratings between 0 and 1.
5  """
6  import pandas as pd
7  from sklearn.preprocessing import MinMaxScaler
8
9  df = pd.read_csv("movies.csv")
10
11 # Remove duplicates
12 df.drop_duplicates(inplace=True)
13
14 # Handle missing values
15 df['rating'].fillna(df['rating'].mean(), inplace=True)
16 df['genre'].fillna('Unknown', inplace=True)
17
18 # Encode categorical variables
19 cat_cols = ['genre', 'language']
20
21 for col in cat_cols:
22     df[col] = df[col].astype('category').cat.codes
23
24 # Scale ratings
25 scaler = MinMaxScaler()
26 df[['rating']] = scaler.fit_transform(df[['rating']])
27
28 print(df.head())
```